

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 01: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE
TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-01

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 The side of a small cube is measured as $x = (10.0 \pm 0.1) \text{ cm}$. What is the maximum estimated percentage error in its calculated volume?

- (A) 1.0% (B) 2.0%
(C) 3.0% (D) 0.1%

Q.2 If the slope of the T^2 versus L graph for a simple pendulum is found to be $4.00 \text{ s}^2/\text{m}$, what is the experimental local value of g (take $\pi = 3.1416$)?

- (A) 9.87 m/s^2 (B) 9.80 m/s^2
(C) 10.0 m/s^2 (D) 9.65 m/s^2

Q.3 A hollow metallic sphere of external radius R and internal radius r has mass M . What is the density of the metal?

- (A) $\frac{3M}{4\pi(R^3 - r^3)}$ (B) $\frac{3M}{4\pi(R^3 + r^3)}$
(C) $\frac{M}{\frac{4}{3}\pi R^3}$ (D) $\frac{M}{4\pi(R - r)^3}$

Q.4 A simple pendulum has time period T on Earth. What is its time period when placed inside an artificial satellite orbiting Earth in free fall?

- (A) T (B) $T/6$
(C) Zero (D) Infinitely large (it does not oscillate)

Q.5 What is the least count of a vernier caliper if 1 main scale division is 0.5 mm and 20 vernier divisions coincide with 19 main scale divisions?

- (A) 0.025 mm (B) 0.05 mm
(C) 0.01 mm (D) 0.1 mm

Q.6 An iceberg floats in seawater (density 1.03 g/cm^3) with pure ice density 0.92 g/cm^3 . What percentage of the iceberg's total volume remains hidden underwater?

- (A) 89.3% (B) 10.7%
(C) 92.0% (D) 80.0%

Q.7 A solid block of mass 120 g has dimensions $4 \text{ cm} \times 5 \text{ cm} \times 6 \text{ cm}$. When dropped into water, what fraction of its volume is submerged?

- (A) 100% (it sinks completely) (B) 100% (it barely floats)
(C) 80% (D) 50%

Q.8 A pendulum clock that keeps correct time at sea level is taken to a high mountain peak where g is 0.2% less. The clock will:

- (A) Gain time by 86.4 s/day (B) Lose time by 86.4 s/day
(C) Lose time by 172.8 s/day (D) Keep identical time

Q.9 The zero mark on the circular thimble of a screw gauge lies 5 divisions above the baseline when closed. It has least count 0.01 mm. The zero error is:

- (A) $+0.05 \text{ mm}$ (B) -0.05 mm
(C) $+0.01 \text{ mm}$ (D) -0.01 mm

Q.10 Which fundamental physical constant has SI units $\text{N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$?

- (A) Universal Gravitational Constant (G) (B) Acceleration due to gravity (g)
(C) Planck's constant (D) Electric permittivity

Q.11 In scientific notation, how is the mass of an electron recorded ($9.109 \times 10^{-31} \text{ kg}$)?

- (A) 4 significant figures (B) 3 significant figures
(C) 31 significant figures (D) 1 significant figure

Q.12 A student measures length of a rod using three instruments: (A) Meter rule, (B) Vernier Calipers, (C) Screw gauge. Which order of precision is correct?

- (A) $A > B > C$ (B) $C > B > A$
(C) $B > C > A$ (D) $A = B = C$

Q.13 What is the dimensional formula of density?

- (A) $[\text{ML}^{-3}]$ (B) $[\text{ML}^{-2}]$
(C) $[\text{MLT}^{-1}]$ (D) $[\text{M}^0 \text{L}^{-3}]$

Q.14 If a solid metal cube of side L expands linearly by 0.1% upon heating, what is the fractional increase in its volume?

- (A) 0.1% (B) 0.2%
(C) 0.3% (D) 0.001%

Q.15 A beaker containing water is placed on a sensitive pan balance. An iron ball suspended from an external stand by a thread is lowered into the water without touching the beaker. The balance reading:

- (A) Remains unchanged (B) Increases by the mass of water displaced
(C) Decreases (D) Becomes zero

Q.16 A simple pendulum of length L has period T . If the acceleration due to gravity increases by 4%, its period approximately:

- (A) Increases by 4% (B) Decreases by 2%
(C) Increases by 2% (D) Decreases by 4%

Q.17 A body floats in liquid A with half its volume submerged, and in liquid B with $\frac{2}{3}$ of its volume submerged. What is the ratio of densities of liquid A to liquid B?

- (A) 3:4 (B) 4:3
(C) 1:2 (D) 3:2

Q.18 Which of the following measurements is recorded with the greatest precision?

- (A) 4.0 m (B) 4.00 m
(C) 4.000 m (D) 40 m

Q.19 When calculating the area of a rectangle of sides 12.2 cm and 3.4 cm, the final answer rounded to proper significant figures is:

- (A) 41.48 cm² (B) 41.5 cm²
(C) 41 cm² (D) 42 cm²

Q.20 A piece of ice floats in a beaker filled to the brim with water. When the ice melts completely, the water level:

- (A) Overflows (B) Drops noticeably
(C) Remains exactly the same (D) First drops then overflows

Q.21 What is the time period of a simple pendulum of length $L = 9.8 \text{ m}$ where $g = 9.8 \text{ m/s}^2$?

- (A) $2\pi \text{ seconds} \approx 6.28 \text{ s}$ (B) 1.0 s
(C) 2.0 s (D) 3.14 s

Q.22 A vernier caliper has 50 divisions on its vernier scale coinciding with 49 mm of main scale. Its least count is:

- (A) 0.02 mm (B) 0.01 mm
(C) 0.05 mm (D) 0.1 mm

Q.23 If density of gold is 19.3 g/cm^3 , what is its density in SI units?

- (A) 193 kg/m^3 (B) 1930 kg/m^3
(C) 19300 kg/m^3 (D) 1.93 kg/m^3

Q.24 A hydrometer stem is calibrated with divisions from top to bottom. The markings indicate:

- (A) Higher densities at the top, lower at the bottom (B) Lower densities at the top, higher densities towards the bottom
(C) Uniform density throughout (D) Temperature rather than density

Q.25 Which experimental measurement has zero systematic instrumental error?

- (A) An uncalibrated ruler (B) A stopwatch with broken spring
(C) An ideal digital counter counting discrete events (e.g., number of full oscillations) (D) A warped measuring tape

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(C)	Volume $V = x^3$. The relative percentage error is $\frac{\Delta V}{V} \times 100 = 3 \times \left(\frac{\Delta x}{x}\right) \times 100 = 3 \times \left(\frac{0.1}{10.0}\right) \times 100 = 3.0\%$.
Q.2	(A)	$\text{Slope} = \frac{4\pi^2 g}{\text{Slope}} \Rightarrow g = \frac{4\pi^2}{\text{Slope}} = \frac{4(3.1416)^2}{4.00} = (3.1416)^2 \approx 9.87 \text{ m/s}^2$.
Q.3	(A)	Volume of metal in hollow sphere $= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(R^3 - r^3)$. Density $\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{3M}{4\pi(R^3 - r^3)}$.
Q.4	(D)	Inside an orbiting satellite, effective gravity $g_{\text{eff}} = 0$ (weightlessness). $T = 2\pi\sqrt{L/g} \rightarrow \infty$; the pendulum does not oscillate.
Q.5	(A)	$\text{LC} = \frac{1}{\text{MSD}} = \frac{0.5 \text{ mm}}{20} = 0.025 \text{ mm}$.
Q.6	(A)	Fraction submerged $= \frac{\rho_{\text{ice}}}{\rho_{\text{seawater}}} = \frac{0.92}{1.03} \approx 0.893 = 89.3\%$.
Q.7	(A)	Volume $= 4 \times 5 \times 6 = 120 \text{ cm}^3$. Density $= \frac{120 \text{ g}}{120 \text{ cm}^3} = 1.00 \text{ g/cm}^3$. Being equal to water density, it remains fully submerged in equilibrium (100%).
Q.8	(B)	Fractional period change $\frac{\Delta T}{T} = -\frac{1}{2} \frac{\Delta g}{g} = +0.1\%$. Time lost per day $= 0.001 \times 86400 \text{ s} = 86.4 \text{ seconds}$.
Q.9	(B)	When the circular scale zero is above the reference line, the error is negative ($-5 \times 0.01 = -0.05 \text{ mm}$) and must be added to observed readings.
Q.10	(A)	From $F = G \frac{m_1 m_2}{r^2} \Rightarrow G = \frac{F r^2}{m_1 m_2}$, unit is $\text{N} \cdot \text{m}^2 / \text{kg}^2$.
Q.11	(A)	All digits in the mantissa (9, 1, 0, 9) are significant; the power of ten does not affect significant figure count (4 significant figures).
Q.12	(B)	Screw gauge (LC 0.01 mm) > Vernier (LC 0.1 mm) > Meter rule (LC 1 mm). Finer least count equals higher precision.
Q.13	(A)	$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{[M]}{[L]^3} = [ML^{-3}]$.
Q.14	(C)	Volumetric expansion is 3 times linear expansion: $\frac{\Delta V}{V} \approx 3 \times \frac{\Delta L}{L} = 3 \times 0.1\% = 0.3\%$.
Q.15	(B)	By Newton's third law, if water exerts an upward buoyant force F_b on the ball, the ball exerts an equal downward force F_b on the water, increasing the scale reading.
Q.16	(B)	Since $T \propto g^{-1/2}$, fractional change $\frac{\Delta T}{T} \approx -\frac{1}{2} \frac{\Delta g}{g} = -\frac{1}{2}(+4\%) = -2\%$.
Q.17	(B)	$\frac{1}{\rho_A} = \frac{2}{3} \rho_B \Rightarrow \frac{\rho_A}{\rho_B} = \frac{2}{3} \times \frac{1}{2} = \frac{4}{3}$.
Q.18	(C)	'4.000 m' measures to the nearest millimeter (0.001 m), representing the smallest least count and highest precision.
Q.19	(C)	$12.2 \times 3.4 = 41.48 \text{ cm}^2$. The least number of significant figures in the data is 2 (in 3.4 cm), so the result is rounded to 2 significant figures: 41 cm^2 .
Q.20	(C)	Floating ice displaces water equal to its own mass. When melted, the resulting water volume exactly equals the submerged volume it previously displaced.
Q.21	(A)	$T = 2\pi\sqrt{9.8 / 9.8} = 2\pi(1) = 2\pi \approx 6.28 \text{ s}$.
Q.22	(A)	$\text{Least count} = \frac{1 \text{ mm}}{50} = 0.02 \text{ mm}$.
Q.23	(C)	$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3 \Rightarrow 19.3 \times 1000 = 19300 \text{ kg/m}^3$.

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.24	(B)	In denser liquids, the hydrometer floats higher (sinks less), so lower numerical readings are at the top and higher readings at the bottom.
Q.25	(C)	Counting discrete whole events (integers) has no instrumental scale error, unlike analog continuous scale instruments.